

AUS & Thai Bank Valuation — Mid-Project Report

Date: 9 July 2026 Prepared for: family review Stage: 2 (residual income valuation model)

This note is a progress update on the bank valuation project. Stage 1 used a quick single-line Gordon Growth screen; we have now replaced it with a proper **5-year residual income model** and a genuine **per-bank cost of capital** built from the individual risks each bank carries. The output is a **fair Price-to-Book (P/B) ratio** for every bank, which we compare against what the market is charging today. The model covers 16 banks — 8 Australian and 8 Thai.

1. How the valuation model works

Banks are balance-sheet businesses: their deposits and borrowings are the raw material they lend out, not just financing. That makes the usual cash-flow / enterprise-value approach a poor fit, so we value the **equity directly** off its book value using **residual income** — the profit a bank earns *above* what its shareholders require.

The single idea behind the model: a bank is worth its book value **plus** the present value of every future dollar it earns over and above its cost of capital. Each forecast year:

- **Residual income** = $(\text{ROE} - \text{cost of equity}) \times \text{opening book value per share}$.
- If a bank earns exactly its cost of capital, it adds nothing above book — it is worth $1.0 \times \text{book}$ (fair $P/B = 1$).
- Only banks earning **more** than their cost of capital are worth a premium to book; those earning less are worth a discount.

We forecast each bank's return on equity (ROE) and dividend payout over 5 years, let them drift toward a long-run 'normal' level, then add a terminal value for everything beyond year 5. Summing today's book value, the 5 years of discounted excess returns, and the terminal value gives the **intrinsic value per share**.

2. The cost of capital — the risks, and how we build it per bank

The 'cost of capital' (cost of equity) is the annual return a shareholder should demand for owning a given bank. A riskier bank must be discounted harder, which lowers its fair P/B . We build it for **each individual equity** using the standard CAPM formula, broken into four separate risk pieces:

Cost of equity = risk-free rate + (beta × equity risk premium) + country risk premium

- **Risk-free rate** — the return on that country's government bonds (the pure time-value of money, no default risk). Australia 4.3%, Thailand 2.5%.
- **Beta** — how much the individual stock amplifies overall market swings (its *market risk*). A beta of 1.0 moves with the market; the big four sit near 0.95–1.10, Macquarie and Judo are higher (1.25–1.30) as more volatile names. **This is the piece that is set bank by bank.**
- **Equity risk premium** — the extra return investors demand for holding shares instead of bonds at all. Australia 5.5%, Thailand 7.0% (an emerging market carries a higher base premium).
- **Country risk premium** — an extra charge for *FX risk, political risk, liquidity risk and general macro uncertainty*. Australia 0.0%; Thailand +3.0%.

A key modelling choice worth calling out: the country risk premium is added as a **constant, parallel spread** — the same +3.0 percentage points is layered onto every Thai bank's discount rate, shifting the

whole Thai group up uniformly rather than singling out individual banks. It is the systematic 'Thailand tax' on required return; bank-specific differences within a country come through beta. The result is that Australian banks land at roughly 9.5%–11.2% and Thai banks at 11.1%–13.2%. That country spread is the single biggest reason Thai banks can trade below their Australian peers for the same profitability — not because they are 'cheaper', but because their cash flows are discounted harder.

Bank	Ctry	Beta	Risk-free	Equity prem.	Country prem.	Cost of equity
Commonwealth Bank (CBA)	AUS	0.95	4.3%	5.5%	0.0%	9.5%
Macquarie Group (MQG)	AUS	1.25	4.3%	5.5%	0.0%	11.2%
ANZ Group (ANZ)	AUS	1.10	4.3%	5.5%	0.0%	10.4%
Judo Bank (JDO)	AUS	1.30	4.3%	5.5%	0.0%	11.5%
Bangkok Bank (BBL)	THA	0.95	2.5%	7.0%	3.0%	12.2%
Kasikornbank (KBANK)	THA	1.00	2.5%	7.0%	3.0%	12.5%
Kiatnakin Phatra (KKP)	THA	1.10	2.5%	7.0%	3.0%	13.2%
Bank of Ayudhya (BAY)	THA	0.80	2.5%	7.0%	3.0%	11.1%

Table 1. A sample of the per-bank cost-of-capital build. Every Thai bank carries the same +3.0% country premium (the parallel spread); beta is what varies name by name.

3. The valuation step — worked example: Commonwealth Bank

Take CBA. Its book value is A\$46.19 per share and it currently earns about a 13.6% return on that book, against a 9.5% cost of capital — so it is a genuine value-creator. Rolling the model forward:

- **Year 1:** earns 13.3% on A\$46.19 book = A\$6.16 EPS; pays out ~78% as dividend, retains the rest, so book grows to A\$47.56. Excess return over its 9.5% cost = **A\$1.76** of residual income (A\$1.60 in today's money).
- **Years 1–5:** the same calculation each year, with ROE fading gently toward its long-run level. The five years of excess returns are worth **A\$6.00** per share today.
- **Beyond year 5 (terminal value):** the remaining excess returns, growing at 3.5% forever and discounted back, are worth **A\$14.23** per share today.

Intrinsic value = A\$46.19 (book) + A\$6.00 (5-yr excess) + A\$14.23 (terminal) = A\$66.41 per share.

4. The P/B estimate step

We then express that intrinsic value as a multiple of book value, because P/B is the natural yardstick for banks and lets us compare all 16 names on one scale regardless of currency or share price:

- **Fair P/B** = intrinsic value ÷ book value = A\$66.41 ÷ A\$46.19 = **1.44×**.
- The market is currently pricing CBA at **3.56×** book.
- **Mispricing** = 1.44 / 3.56 – 1 = **–60%** — the model views CBA as richly priced versus the excess returns it can be expected to generate.

Running this for all 16 banks gives the picture in the two charts below.

5. Fair P/B vs actual P/B — the whole universe

Each dot is a bank. The dashed line is where fair value equals the market price. Dots **below** the line trade above their model fair value (expensive); dots **above** the line are cheap.

Residual income fair P/B vs actual P/B (above line = undervalued)

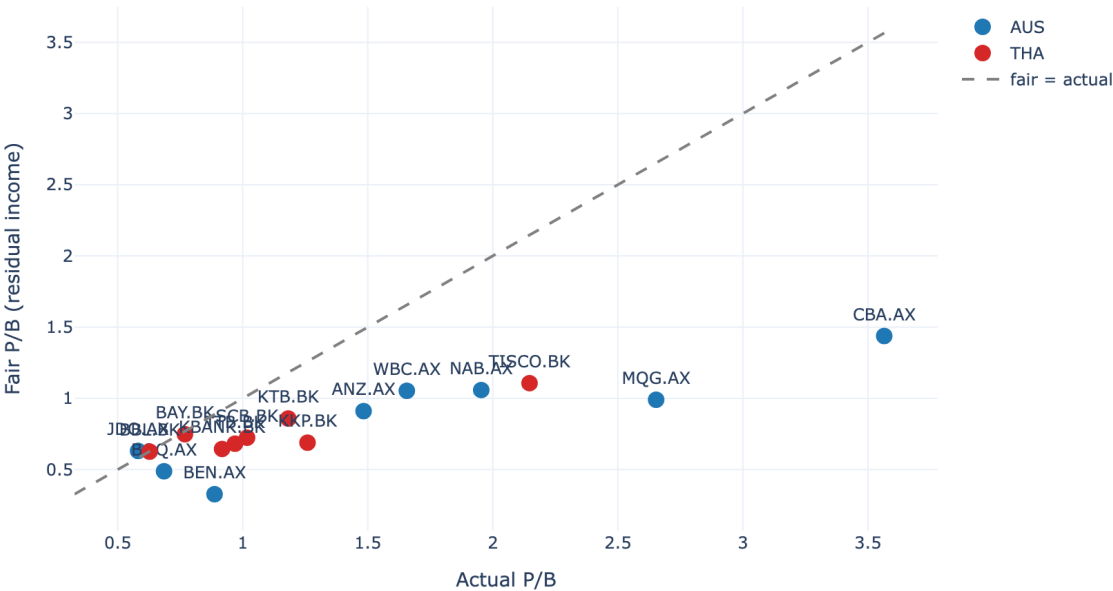


Figure 1. Residual-income fair P/B (vertical) against today's market P/B (horizontal). Almost every bank sits below the parity line — on these cost-of-capital settings the market is paying more than the model's excess-return value, most extremely for the high-quality Australian names (CBA, Macquarie).

6. Fair P/B by scenario vs current market P/B

This view stress-tests each bank across a **bear / base / bull** case (we flex ROE, cost of capital and long-run growth together) and overlays where the market is pricing the bank right now. The black dot is the base-case fair P/B, the whiskers stretch from bear to bull, and the red x is today's market P/B.

Fair P/B by scenario (whiskers = bear..bull) vs current market P/B

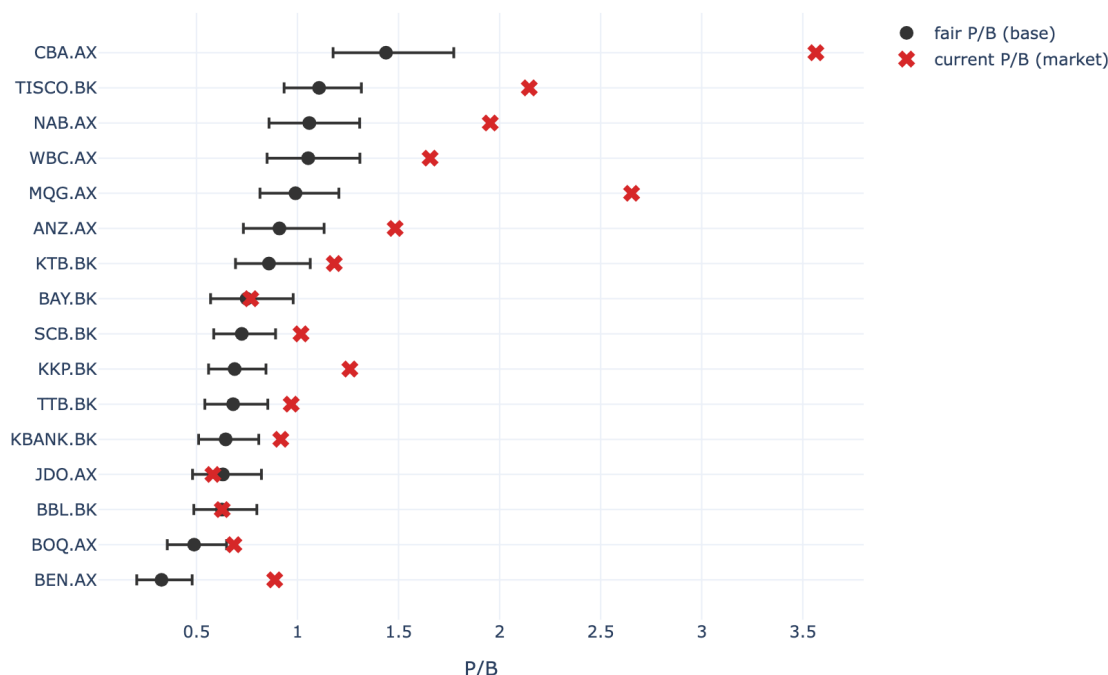


Figure 2. Where a red × sits to the right of a bank's whisker, the market is pricing it above even the optimistic (bull) fair value. For most banks the × is at or beyond the right whisker — the 'overvalued' read is not fragile to the assumptions.

7. Main growth assumptions (and that they can change)

The forecasts are mechanical and fully editable in one place — **config/assumptions.yaml**. Nothing in the numbers above is hard-wired; these are the current settings and can be revised as our views change:

Assumption	Australia	Thailand	What it does
Long-run (terminal) growth	3.5%	4.0%	Growth of excess returns beyond year 5, forever
Long-run 'normal' ROE	11.0%	10.5%	Where each bank's ROE drifts toward
Long-run payout ratio	75%	55%	Share of earnings paid as dividends long-term
Mean-reversion weight	0.60	0.60	How hard a bank pulls to the country norm vs its own current level
Forecast horizon	5 years	5 years	Explicit years before the terminal value kicks in

Table 2. Current growth / long-run assumptions. Each bank fades from its own present ROE and payout toward these country anchors over the 5-year forecast.

Where things stand. The model now values banks the way a bank analyst would — off book value and excess returns, with a real per-bank cost of capital that makes Australia's and Thailand's different risk explicit. On today's settings almost every bank screens as fully priced, and that conclusion holds up across the bull/bear range. Next steps: replace the placeholder betas with measured ones and back-test the fair-P/B signal against actual returns.